



Fairness Score and process standardization: framework for fairness certification in artificial intelligence systems

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Abstract

Decisions made by various artificial intelligence (AI) systems greatly influence our day-to-day lives. With the increasing use of AI systems, it becomes crucial to know that they are fair, identify the underlying biases in their decision-making, and create a standardized framework to ascertain their fairness. Biases in AI systems lead to unintended ethical, social and even legal issues. In this paper, we propose a novel and versatile Fairness Score and Bias Index for measuring the fairness of a supervised learning AI system. We also propose a standard operating procedure (SOP) for issuing Fairness Certification for such data-driven applications. Fairness Score and audit process standardization will ensure quality, reduce ambiguity, enable comparison and improve the trustworthiness of the AI systems. It will also provide a framework to operationalize the concept of fairness and facilitate the commercial deployment of such systems. Furthermore, a Fairness Certificate issued by a designated third-party auditing agency following the standardized process would boost the conviction of the organizations in the AI systems that they intend to deploy. The Bias Index proposed in this paper reveals comparative bias amongst the various protected attributes within the dataset, and the Fairness Score provides overall fairness with an ideal value of 1. To substantiate the proposed framework, we iteratively train a model on biased and unbiased data using multiple labelled datasets and check that the Fairness Score, Bias Indexes and the proposed process correctly identify the biases and judge the fairness.

Keywords Fairness Score · Bias Index · Standardization · Artificial intelligence · Ethics · Audit · SOP

1 Introduction

Artificial intelligence (AI) services are used widely in various fields and contexts such as finance, education, medicine, and law. Supervised learning AI systems use large datasets for training and several algorithms to enable accurate and fast decision-making. These systems make decisions based on their learnings from the training data and algorithms. The systems are improving continuously, becoming more complex and automated. As such, manual supervision and validation of their outcomes are reducing. Thus it is pertinent to ask some fundamental questions, such as (1) how the system reaches the outcome? (2) how often the system's

outcomes are wrong (3) is the system being unfair to particular demography? [1].

Over the centuries, human psychology has shown partisanship based on race, gender, religion, age, etc. The general belief in society that the decisions made using AI systems will be fair may not be correct. These systems learn from datasets consisting of real-world events and observations. The biases or prejudices present in the real world seep into the systems through various steps in model building. Hence, this discrimination can be observed in the results and decisions made by the AI systems. The discrimination and discriminatory decisions can be termed as biases and biased decisions, respectively. Biased decisions hinder the rights and sentiments of the unprivileged group and may cause a negative social impact. They instigate a lack of trust and may increase the social and financial disparities instead of decreasing them. They may create a feeling of being left out in the unprivileged groups, thus violating the laws to provide equal opportunity and treatment to all citizens. Biased AI systems raise various ethical, social as well as

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legal issues. Reference [2] discusses ethical implications of bias in machine learning.

Thus, it is necessary to detect biases in AI systems and then mitigate them so that these systems are fair. However, how do the users know whether a decision involving them is made using a fair system or a biased system? Can the developing organization's claim to fairness be taken at face value? Are there metrics/ methodologies to verify the claim? Can some audit processes be standardized such that neutral certifying agencies can audit and certify AI systems for fairness? We answer some of these questions in this paper. A single audit framework may not be equally applicable to all categories of AI systems. This paper focuses on assessing the fairness of supervised learning AI systems and enabling their third-party audits and certification. Recommending corrective actions for AI systems not meeting the qualifying criteria is beyond the scope of this paper.

Our contributions:

1. We propose a Fairness score to measure the overall fairness of a supervised learning AI system. We also propose Bias Index to measure the bias of each protected attribute in a dataset. The overall Fairness Score is derived from the Bias Index for each protected attribute.
2. We propose a standard operating procedure (SOP) for issuing Fairness Certification for data-driven AI systems that would provide a framework to operationalize the concept of fairness and facilitate commercial deployment of such systems.
3. We explain the need for third-party Fairness Certification for such systems.
4. We define the desired characteristics of metrics to detect bias at different phases of such AI systems.
5. We propose the metrics out of a plethora of such metrics sufficient to detect bias.

Different countries, cultures, or even societal sectors focus on different values and their views on fairness might differ. The Fairness Score, Bias Index and the SOP proposed in this paper are flexible to accommodate different perspectives as they allow choice in the selection of (i) metrics, (ii) privileged and unprivileged classes, and (iii) the criteria for favourable outcomes. Still, they capture the degree of bias in different supervised learning AI systems on the same scale, thus, making the Fairness Score and Bias Index versatile.

Methodology

We review the various causes of bias in AI systems and different ways to define fairness. Then, we shortlist the metrics that properly detect bias in a training dataset and a trained model. Accordingly, we propose the Fairness Score, Bias Index, and an SOP for detecting bias. Finally,

we establish our proposals by applying them to a model, which is trained multiple times using the following popular open datasets one by one:

1. UCI Adult dataset (four scenarios):
 - (a) Original dataset.
 - (b) Dataset after balancing for gender.
 - (c) Dataset after balancing for race.
 - (d) Dataset after balancing for marital status.
2. German Credit dataset (two scenarios):
 - (a) Original dataset.
 - (b) Dataset after balancing for gender.
3. Health Insurance Cross-sell Prediction dataset (one scenario)
 - (a) Original dataset.

Related work

Various researchers have worked in areas related to biases in AI systems, concepts of fairness, and metrics to measure fairness, as elaborated in subsequent sections. Most of these works are related to theoretical or mathematical aspects. Our proposal is different as we focus on the practical implementation aspects aiming to standardize the audit procedure to check the fairness of an AI system to enhance its trustworthiness. While most researchers have proposed different metrics to check for biases, we consider these metrics and present an integrated Fairness Score and Bias Index. Our proposal fills the gap regarding the ethical requirement of neutral verification of fairness of AI systems. Our work would be found beneficial by organizations developing AI systems as well as those deploying such systems. It would also facilitate benchmarking and independent third-party audits for fairness. This would also be valuable to policymakers and legislation in evaluating the fairness of an AI system.

2 Need for fairness certification

Bias creates an unwanted imbalance between different groups. It gives an advantage to the people belonging to one particular demography irrespective of other vital parameters that value that decision-making process. Apart from this, it also affects the business and its profits based on the predictive results of the system. Therefore, ensuring the trustworthiness of AI systems is not only an ethical requirement but also a business requirement [3]. Moreover, AI bias could negatively affect every sector as AI systems become widespread [4].

Previous studies have highlighted cases where biases in AI-driven systems have caused unwanted impacts and

results. For example, Amazon developed an online AI-based recruitment system where the candidates were shortlisted based on their resumes. By 2015, it came to the company's attention that ratings were biased in favour of male applicants for technical job titles like software developers and architects. On investigation, it was found that the data used for training the system was having gender bias as it consisted of resumes of mostly male employees in reflection to the then actual hiring trend in the company [5]. Research on online freelance marketplace based on evidence from TaskRabbit and Fiverr revealed that social feedback on these sites often has a significant statistical relationship with perceived gender and race [6]. Various facial recognition systems have biases and give accurate detection results only for people of a certain race and often incorrect results for the other demographic groups [7]. The US healthcare system was observed to be racially biased and preferred treating white patients over the ones with African origins [8]. The Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) score, used in States such as California and Florida, was reported to be biased against African origin inmates according to certain measures of fairness [9]. Princeton research showed that off-the-shelf AI software was biased in associating words like 'woman' and 'girl' with arts as compared to science or math [10]. Studies showed that online search queries were more likely to return ads from services related to arrest records when names of a particular race were searched than when names of the other race were searched [11, 12]. Various search engines and recommender systems had race and gender discrimination [13]. Biases in expert systems can also lead to the false detection of disease, wrong prescription of medical tests and more visits to the hospital for routine check-ups [14].

These and many similar kinds of research indicate the negative impact of biases in AI systems and the need for ensuring fairness in such systems. These raise ethical issues about various such decision-making systems and cause reputational damage against the organizations deploying them. Biased systems reduce opportunities for the organizations as well as customers and other populations that are affected by their decision. As AI systems get embedded in day-to-day applications and essential services such as healthcare, judicial, and many more, it becomes pertinent for users to know that the system they are using is fair. Small service providers to large corporations and government organizations use AI/ Machine Learning (ML)-based applications to provide services. However, everyone does not have the resources or technical know-how to test the fairness of the AI systems they are deploying. The AI application could be an off-the-shelf product or a customized product developed specifically for that end-user. There is, therefore, a need for a standard testing procedure to test the AI application for biases. A Fairness Certificate issued by a neutral certifying

authority based on such standard testing procedure would assure the users of the fairness of the AI/ ML system.

The standard testing procedure should protect proprietary AI system developers, privacy, and security of the training dataset [15]. It should comply with the existing national and international regulations [16]. Finally, it should not only protect the rights of individuals but also help the competition to grow and provide the best services [17].

3 Causes of bias in AI systems

Biases can enter an AI system at any point in its cycle. Empirical findings have shown that data-driven methods can unintentionally encode existing human biases and introduce new ones [18]. According to [19], three causes of biases are:

Bias in modelling:

These biases correspond to the algorithmic biases, caused due to the algorithms defined for model functioning and decisive actions. Algorithms often reflect biases of the developers and hence the society at large [20]. Also, they do it at a potentially massive scale and without due oversight [21].

Bias in training:

The systems learn based on the data used to train them. The biases existing in society are often manifested in the data used by AI algorithms and can be modelled and formally defined [22]. The data may represent the ideologies of only one group of the population or specific cases, which introduces prejudices or biases in the decisions.

Bias in usage:

These biases arise when the models or systems are used for a purpose not defined for them. The AI systems designed for making decisions for a specific group or section provide biased results when used for any other group. Misinterpretation of the outcomes of the system also causes wrong and indeliberate actions or biases. Biases also creep in when the systems are used for the intended purpose but one specific class has more use instances while the AI system keeps learning from its usage.

Reference [23] also describes similar causes of biases.

4 How to define fairness

While we create a certification framework and identify fairness metrics, it is necessary to define fairness. AI fairness is a rapidly growing topic of inquiry [24]. References [25–27] proposed multiple definitions of fairness. Reference [28] proposed some definitions for formalizing fairness from

other disciplines. In some cases, defining fairness comes as a legal compulsion, for example, lending money through The Equal Credit Opportunity Act in the United States [29].

We can broadly define fairness as the state of being impartial towards every individual and group involved. However, fairness can be perceived differently by different people and contexts [30]. Thus, it is not easy to have a single definition of fairness for all AI systems.

Various researchers have come up with their definitions for fairness in AI systems. The outcomes would differ depending on the difference in the aspects of fairness focused on by different definitions of fairness [4].

To apply fairness well, one should understand it within a specific context. Guiding questions to determine fairness could be: Are there particular groups that the algorithm/system may advantage or disadvantage in that context? In case of uncertainty or errors in the AI system, what is the potential adverse effect to different groups [31]?

Individual fairness implies that if two individuals differ only in protected attributes, such as gender or race, they should receive similar outcomes. There is individual discrimination if they receive different outcomes [32]. A distance metric may be used to measure the similarity between the two individuals [33].

Group fairness involves dividing the data into different categories based on the protected attributes and ensuring that the outcomes are similar for all the groups. Group fairness averages out the data of individuals in a group, so it may or may not represent the correct picture.

Other than individual and group fairness, fairness definitions can also be classified using the statistical approach used: definitions based on the predicted outcome; definitions based on predicted and actual outcomes; and definitions based on predicted probabilities and actual outcomes [25].

Similarity-based definitions use non-sensitive attributes also, unlike the above fairness definitions.

Counterfactual fairness uses plotting a graph to check whether the outcome-defining attributes are correlated to the sensitive attributes [22, 34].

These definitions are independent of each other, and multiple definitions can apply to an AI system simultaneously. It is also important to note that, for an AI system/ dataset to be fair, it does not mean that it should always have equal representation for all the classes across the different demographics. Equality should be in sync with the ground reality.

While many formal mathematical definitions of fairness have been proposed, formalizing fairness is still a problem. These definitions have statistical limitations, and there are adverse effects of enforcing such fairness measures on group well-being [35]. Thus, there might be systems that do not meet some definitions but still are fair. It is, therefore, essential to understand the context of the AI system before judging it for fairness. Obviously, fairness is a multi-disciplinary

topic. Reference [36] provides a survey of fairness analysis in multiple disciplines.

In the subsequent sections, we define the features of a good fairness metric and discuss some of the fairness metrics along with their mathematical representations.

5 Desired characteristics of fairness metrics

We need to measure a data-driven AI system against some fairness criteria or metrics to evaluate if the system is fair [37]. Unwanted biases in training data, because of either prejudice in labels or under-/ over-sampling, can be checked using statistical tests on datasets or models [38, 39].

While the fairness definitions and computation of fairness metrics are explicit, applying inflexible thresholds on such metrics is not practical, as interpreting the numbers is often subjective. As such, we should decide the thresholds considering the constraints in view. Before computing these metrics, it is crucial to define the characteristics of a good fairness metric.

A good fairness metric should be clear in terms of its mathematical and theoretical significance and also in terms of features, outcomes, and situations it takes into consideration during evaluation. There should be clear guidelines on the metric review to draw valuable inferences about the biases in the system. Its constraints, drawbacks, and advantages should be known to enable its selection according to the application of the system.

A good fairness metric is expected to be quantitatively and accurately measurable to calculate the bias present in the AI system. Furthermore, it should assist in coming up with effective mitigation processes for the biases and setting up fairness benchmarking guidelines for an AI system.

An excellent metric should be universally accepted for the systems used in various parts of the world. This is required to avoid discrepancies in the fairness definitions or guidelines for a system in various geographies and to establish uniform justice among people around the globe.

As mentioned in the previous section, the context of fairness is domain-dependent and there may be an inherent trade-off in aggregating multiple metrics in different use cases [40, 41].

6 Standard fairness metrics

Here, we have explained some of the standard fairness metrics which abide by the above characteristics and are accepted widely as correctly capturing biases. The various notations used in their mathematical representations are:

Y : training label

Y' : predicted label

$S \in \{0, 1\}$: binary indicator of the protected class

1 indicates the privileged group

0 indicates an unprivileged group

6.1 Demographic parity

Demographic parity is a measure of whether the probabilities of favourable outcomes for privileged and unprivileged groups of data in a protected attribute are the same.

It is independent of the prediction/ system outcome, making it simpler and easier to implement. However, a drawback to this metric is that it does not consider the genuine real-world differences between the privileged and unprivileged groups. Thus, this metric will be inappropriate if the reason for the difference in probabilities of favourable outcomes for different groups in a protected attribute is legitimate, for example, distribution of breast cancer probability based on gender [42]. For an unbiased system, it can be mathematically represented as:

$$P(Y = 1 | S = 0) = P(Y = 1 | S = 1). \quad (1)$$

In case of bias, the two sides of the equation will not be equal. This metric can be calculated in two different ways:

6.1.1 Statistical parity difference

Statistical parity difference is the difference of the probabilities of favourable outcomes for unprivileged and privileged classes in a protected attribute and represented as:

$$P(Y = 1 | S = 0) - P(Y = 1 | S = 1). \quad (2)$$

For an unbiased system, its value should be zero.

6.1.2 Disparate impact

Disparate impact is the ratio of the probabilities of favorable outcomes for unprivileged and privileged classes in a protected attribute [43] and represented as:

$$\frac{P(Y = 1 | S = 0)}{P(Y = 1 | S = 1)}. \quad (3)$$

For an unbiased system, its value should be one.

6.2 Equal opportunity

Equal opportunity measures the equality of the true positive rates, i.e. equality of recall value for the privileged and unprivileged groups. It simply means that every group

should have equal opportunities to be successful. It is also called positive rate parity. Mathematically, it can be written as [44]:

$$P(Y' = 1 | S = 1, Y = 1) - P(Y' = 1 | S = 0, Y = 1). \quad (4)$$

Its disadvantage is that it requires large labelled historical data [45].

6.3 Equal mis-opportunity

According to equal mis-opportunity, the accuracy measured as false positive rate should be the same for privileged and unprivileged groups. It is also known as predictive equality [46]. Mathematically, it can be written as [42]:

$$P(Y' = 1 | S = 1, Y = 0) - P(Y' = 1 | S = 0, Y = 0). \quad (5)$$

6.4 Average odds

Average odds is the combination of equal opportunity and equal mis-opportunity metrics. It measures whether the true positive and the false positive rates are the same for privileged and unprivileged groups. It simply means that the favourable outcome is independent of the protected feature. Mathematically, it can be written as:

$$(P(Y' = 1 | S = 0, Y = y) - P(Y' = 1 | S = 1, Y = y))/2, \quad (6)$$

where

$$y \in \{0, 1\}$$

It tries to ensure high accuracy in all demographics [44]. It can be preferred when discrimination can be accepted as long as it is justified by actual trustable data [47].

6.5 Distance metrics

Distance metrics are used to calculate the distance between two points. In terms of fairness metrics, these metrics can be used to evaluate the similarity between two data points. These can also be used to compare two datasets, for example, for measuring the change between an original dataset and the dataset after distortion/ bias mitigation.

Some familiar distance metrics are Euclidean distance, Manhattan distance, and the more generalized Minkowski distance.

One disadvantage of these metrics is that large valued variables might influence results. Also, they are not suitable for image classification or document classification [48]. Finally, these metrics are also not suitable to measure similarities between groups within the same dataset.

Some researchers have suggested using distance metrics to measure bias while focusing on using these metrics to determine individual fairness. In our experiment, we consider group fairness and hence have not used the distance metrics to calculate bias. However, the fairness certification framework proposed in this paper can be extended for individual fairness also.

7 Proposed framework for fairness certification

7.1 Fairness Score and Bias Index

Different users might use different metrics to check the fairness of an AI system. Hence, it is crucial to standardise the bias measurement on a linear scale so that a uniform scale can be used to assess fairness and enable the comparison of different AI systems. Therefore, we introduce Bias Index for each protected attribute and Fairness Score for the overall system as the standard benchmarks for measuring fairness.

Bias Index is defined for each protected attribute in the system as follows:

$$BI_i = \sqrt{\frac{\sum_{j=1}^n (M_{ij} - M'_j)^2}{n}}, \quad (7)$$

where

i : number of the protected attribute

j : number of the fairness metric

n : total number of fairness metrics used

m : total number of protected attributes considered in the AI system

M_{ij} : value of the j th fairness metric for the i th protected attribute

M'_j : ideal value of the j th fairness metric
i.e. 0 for difference metrics and 1 for ratio metrics

Fairness Score is defined for the AI system as follows:

$$FS = 1 - \sqrt{\frac{\sum_{i=1}^m (BI_i)^2}{m}} \quad (8)$$

Substituting the equation for BI, we get,

$$FS = 1 - \sqrt{\frac{\sum_{i=1}^m \sum_{j=1}^n (M_{ij} - M'_j)^2}{mn}} \quad (9)$$

While Bias Index corresponds to the degree of bias for a particular protected attribute in a dataset or model, Fairness Score corresponds to the degree of fairness in the entire

model, considering all the protected attributes together. For a fair system, the Bias Index for each protected attribute should be zero, and the Fairness Score should be one. AI systems may have more than one protected attribute, and so there would be as many Bias Indexes as the number of protected attributes, but there will be only one Fairness Score for the model. The AI system might be biased for some of the protected attributes and fair for the other protected attributes, which the corresponding Bias Indexes would reflect; the Fairness Score, on the other hand, would reflect the overall fairness.

7.2 Proposed standard operating procedure for fairness certification

Considering the desired characteristics of the fairness metrics and their applicability at different stages of model building, we propose a standard operating procedure for issuing Fairness Certificate for data-driven AI Systems as follows:

1. Identify the protected attributes in the dataset used in the AI system.
2. Identify the privileged and unprivileged classes for each attribute in the dataset.
3. Define the tolerance band for each attribute. Ideally, it should remain fixed for proper benchmarking.
4. Check the training dataset for fairness. Compute values of statistical parity difference and disparate impact (or any other metric considered appropriate) for each protected attribute of the training dataset. Check if the values fall in the tolerance band.
5. Train the AI system using the training dataset.
6. Run the system on a test dataset and tally the outcomes.
7. Compute values of equal opportunity difference, equal mis-opportunity difference, and average odds difference (or any other metric considered appropriate) for each protected attribute of the outcome. Check if the values fall in the tolerance band.
8. Plot the various metrics on a graph and see if all the metrics are within the tolerance band.
9. Calculate the Bias Index for each protected attribute and the Fairness Score for the overall AI system.
10. Based on the above observations, the fairness certification can be issued.
11. In a real-world scenario, the developers might not share the training dataset with the certifying agency. In such cases, steps 4 and 5 have to be skipped. Therefore, the test dataset should be close to unbiased for all the protected attributes. Any deviation of the metrics computed on the outcomes from the mean position would indicate bias in the system. In such cases, the certificate issued by the certifying agency should mention that the statistical parity and disparate impact have not

been checked because of the non-availability of the training dataset. Furthermore, a legal framework could be considered where depending on the requirements and use case, the developers can be mandated to share their training datasets with the certifying agency for certification.

12. As the system is operationalized and keeps learning from real-world data, it is necessary to check periodically for any biases introduced after the initial tests. As such, periodic recertification is recommended.

8 Experiment

We evaluated our proposed framework on seven different datasets related to the popular open datasets: UCI Adult dataset (four scenarios), German Credit dataset (two scenarios), and Health Insurance dataset. For each dataset, we performed the following steps:

We identified the protected attributes and also the privileged and unprivileged classes for each such attribute. Then, we cleaned the dataset by removing special characters, dropping some skewed attributes, and transforming the categorical attributes using one-hot encoding. We randomly divided the dataset into 85% training set and 15% test set. We used RandomForestClassifier to train the model and make predictions. We calculated the following fairness metrics for each protected class: statistical parity difference (SPD) and disparate impact (DI) for training dataset; equal opportunity difference (EOD), equal mis-opportunity difference (EMOD), and average odds difference (AOD) for test set predictions using the trained model.

We used visualization tools to plot these observations. We finally calculated the Bias Index for each protected attribute and the Fairness Score for the model. We repeated these for all seven cases as mentioned above.

8.1 UCI Adult dataset

The UCI Adult dataset is also known as the Census Income dataset. The data has been extracted from the 1994 Census dataset. It is used to predict whether an individual's income exceeds \$50k per annum or not. The dataset contains attributes like age, gender, marital status, occupation, among others. This dataset is widely known for its gender and racial bias. More percentage of records belonging to men have high income as compared to that of women. A model trained using this dataset is more likely to predict a higher income for a male candidate compared to a female candidate with similar attributes/ skills. Similarly, bias in the dataset for

race and marital-status leads to unfairness amongst the equally placed candidates.

As part of pre-processing, we reclassified the classes in the marital-status attribute as following:

```
Married = {'Married-civ-spouse',
           'Married-AF-spouse'}
Unmarried = {'Widowed',
             'Never-married', 'Divorced',
             'Separated', 'Married-spouse-absent'}
```

For our experiment, we considered the following configurations:

Protected attributes: gender, race, marital-status

Privileged class: Male (gender), White (race), Married (marital-status)

Unprivileged class: Female (gender),

Other than White (race),

Unmarried (marital-status)

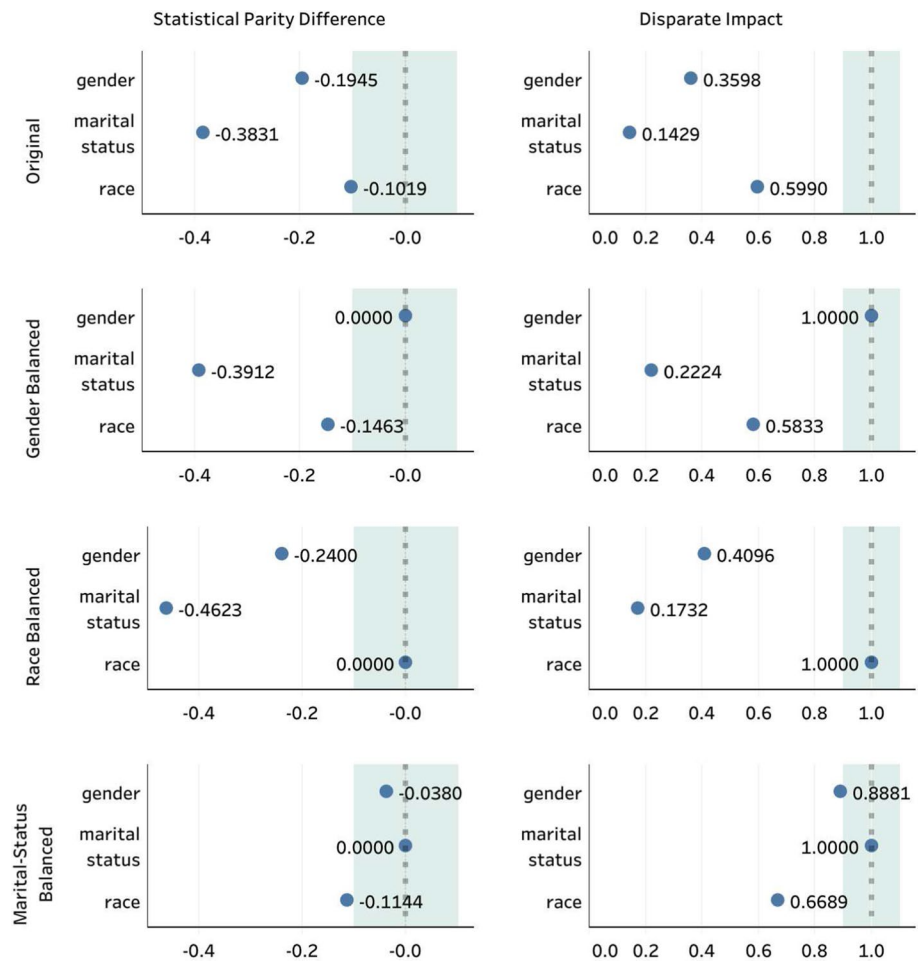
Favourable outcome: Income > \$50k/yr.

We included marital status as a protected field because, on analysis of the UCI Adult dataset, it showed a clear bias towards married individuals compared to individuals not in a relationship. One possible reason could be the belief at that time that married individuals were considered more stable and reliable employees.

We did the first iteration of model training and bias calculation on the original (biased) dataset. For the next iteration, we mitigated bias in the race attribute by balancing the percentage of favourable outcomes for both the privileged and unprivileged classes. We used over-sampling (SMOTENC) and random undersampling. Then, we trained a fresh model on this resampled dataset and calculated a new set of fairness metrics for each protected attribute. We repeated the same steps two more times, balancing gender one time and marital status the other.

Figure 1 shows statistical parity difference and disparate impact calculated on the pre-trained dataset for all four iterations. Figure 2 shows equal opportunity difference, equal mis-opportunity difference, and average odds difference calculated on the model predictions on the test dataset. As can be seen, the original dataset has a bias for all the protected attributes. Furthermore, in subsequent iterations, when we calculate metrics by resampling for each protected attribute one by one, the values start falling in the acceptable band for that attribute. We also observe that while resampling for a protected attribute reduces the bias in that attribute, it also tends to reduce the bias for other attributes.

Fig. 1 Statistical parity difference and disparate impact values for the pre-training datasets for the four scenarios of the UCI Adult dataset



8.2 German Credit dataset

Each record in the German Credit dataset represents an individual classified as either good or bad credit risk. The dataset contains attributes like Sex, Housing, Purpose, Duration among others. This dataset helps to predict if a customer will pay back the loan. This dataset is biased as it tends to classify men as safe borrowers compared to women, leading to inequality in the distribution of funds. A gender bias in the dataset may deprive some female credit seekers of getting a loan.

For our experiment, we considered the following configurations:

Protected attributes : Sex

Privileged class : male (Sex)

Unprivileged class : female (Sex)

Favourable outcome : good (Risk)

Figure 3 shows the fairness metrics calculated on the original dataset and the dataset after mitigating bias in the

protected attribute. The metrics indicate some gender bias in the original dataset that gets addressed in the gender-balanced dataset.

8.3 Health Insurance Cross-sell Prediction dataset

Health Insurance Cross-sell Prediction dataset is an insurance company's dataset of customers to whom the company provided health insurance in the last one year. The dataset is used to predict the customers interested in buying the company's vehicle insurance. The dataset contains attributes like DrivingLicense, VehicleAge, VehicleDamage, AnnualPremium among others.

For our experiment, we considered the following configurations:

Protected attributes : Gender

Privileged class : Male (Gender)

Unprivileged class : Female (Gender)

Favourable outcome : 1 (Response)

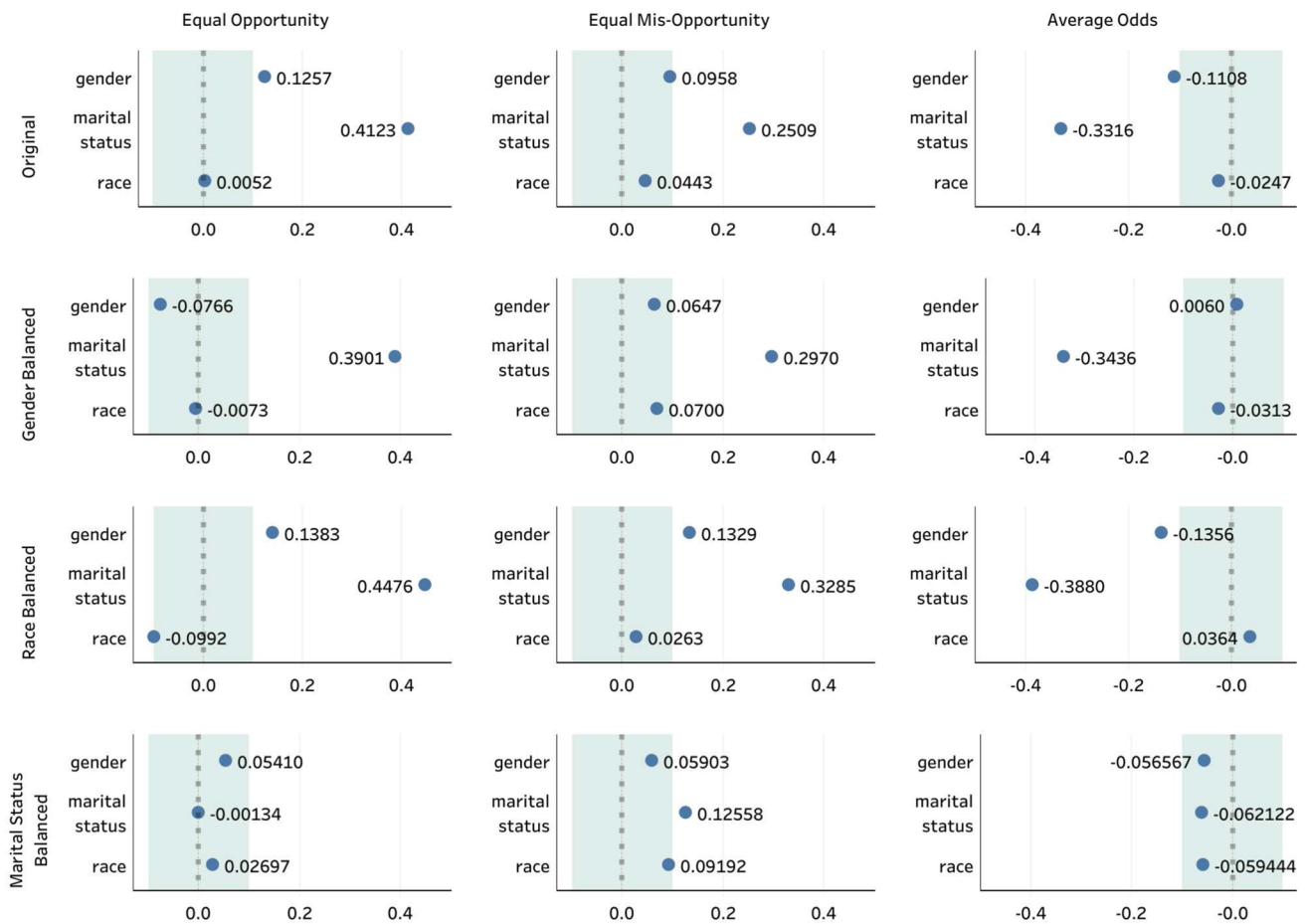


Fig. 2 Equal opportunity, equal mis-opportunity and average odds values for the predictions of the test datasets for the four scenarios of the UCI Adult dataset

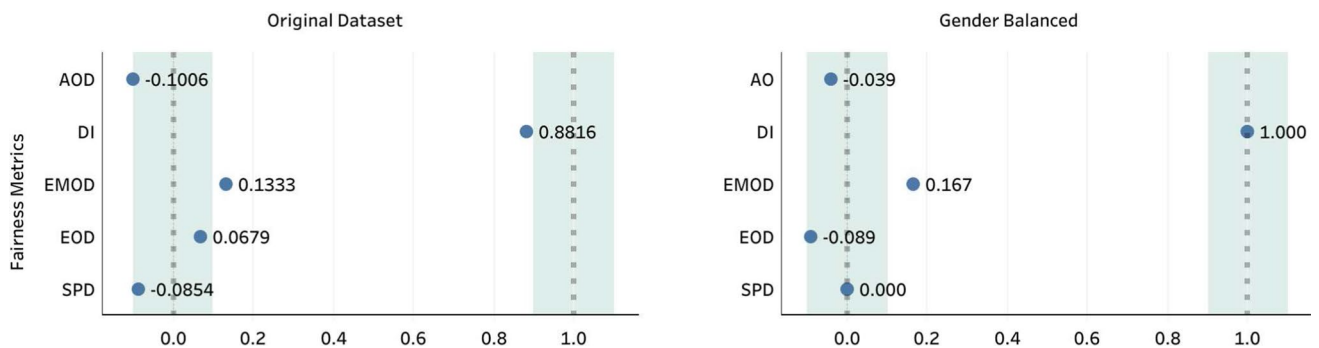


Fig. 3 Fairness metrics for the original (left) and gender-balanced (right) German Credit dataset

Figure 4 shows the fairness metrics calculated on the original dataset. The metrics, except disparate impact, indicate that the dataset is free for bias for the protected attribute.

Fairness Scores calculated for the datasets, Bias Indexes and various fairness metrics calculated for the protected attributes for all the seven cases are tabulated in Table 1.

9 Discussion

Analysis of the calculated values and their graphs related to the experiment are as follows:

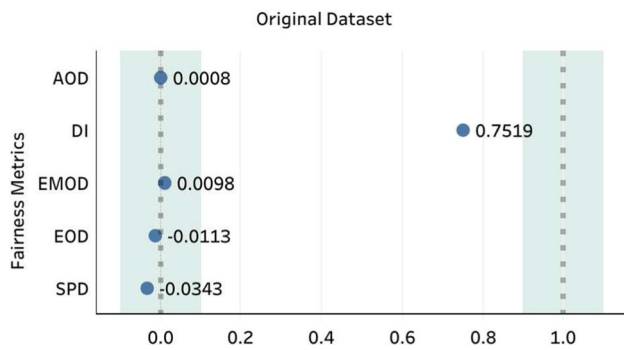


Fig. 4 Fairness metrics for the Health Insurance cross-sell prediction dataset

1. The training datasets used for various AI systems are not perfect, so the AI systems are likely to be prejudiced and may not be fair in all aspects. Biases may also be there because of imperfections in the algorithms in the model. Our experiment indicates that different metrics are required to check for biases in the pre-processed training dataset and the outcomes. For the former case, we have used statistical parity difference and disparate impact. For the latter case, we have used equal opportunity difference, equal misopportunity difference, and average odds difference.
2. A single metric for each part (pre-processed training dataset and outcome being two parts) might not correctly identify bias in all cases, so using a combination of metrics is suggested for each part. We have used two metrics for the pre-processed training dataset and three metrics for the post-training outcomes.
3. Different metrics may be used in other AI systems, but the Fairness Score and Bias Indexes will still be relevant as they are flexible and can compare various supervised learning AI systems.
4. The training dataset might have multiple attributes that might affect the fairness of the system. It is necessary to check for fairness for each protected attribute. In our experiment, we checked for fairness for race, gender, and marital status in the UCI Adult dataset and gender in the other two datasets.
5. Testing all the protected attributes combined is difficult, whereas checking for each attribute separately is practically implementable.
6. Our experiment confirms bias in the protected attributes of race, gender, and marital status in the UCI Adult dataset and gender in the German Credit dataset. Furthermore, no significant bias is observed for the protected attribute of gender in the Health Insurance Cross-sell Prediction dataset. Previous studies validate these results [49].

Table 1 Fairness Scores for the datasets, Bias Indexes and various fairness metrics calculated for the protected attributes for all the seven cases

Protected attribute	SPD	DI	EOD	EMOD	AOD	Bias Index	Fairness Score
UCI adult dataset—original dataset							
Race	-0.1019	0.5990	0.0052	0.0443	-0.0247	0.1864	0.6457
Gender	-0.1945	0.3598	0.1257	0.0958	-0.1108	0.3114	
Marital Status	-0.3831	0.1429	0.4123	0.2509	-0.3316	0.4948	
UCI adult dataset—gender balanced							
Race	-0.1463	0.5833	-0.0073	0.0700	0.0313	0.2005	0.7025
Gender	0.0000	1.0000	-0.0766	0.0647	0.0060	0.0449	
Marital Status	-0.3912	0.2224	0.3901	0.2970	-0.3436	0.4725	
UCI adult dataset—race balanced							
Race	0.0000	1.0000	-0.0992	0.0263	0.0364	0.0487	0.6508
Gender	-0.2400	0.4096	0.1383	0.1329	-0.1356	0.3038	
Marital Status	-0.4623	0.1732	0.4476	0.3285	-0.3880	0.5208	
UCI adult dataset—marital status balanced							
Race	-0.1144	0.6689	0.0270	0.0919	-0.0594	0.1646	0.8909
Gender	0.0380	0.8881	0.0541	0.0590	-0.0566	0.0687	
Marital Status	0.0000	1.0000	-0.0013	0.1256	-0.0621	0.0627	
German credit dataset—original dataset							
Sex	-0.0854	0.8816	0.0679	0.1333	-0.1006	0.1037	0.8963
German credit dataset—sex balanced							
Sex	0.0000	1.0000	-0.0890	0.1670	-0.0390	0.0864	0.9136
Health insurance dataset—original dataset							
Gender	-0.0343	0.7519	-0.0113	0.0098	0.0008	0.1122	0.8878

7. The graphical representation also shows the relative degree of bias for one attribute compared to others. The UCI Adult dataset has a maximum bias in the case of marital status.
8. When bias for one attribute is removed, the impact on other protected attributes can also be checked from this graphical representation. It is observed that when bias for marital status was removed, the biases for the other two protected attributes also got reduced.
9. As having a perfect unbiased system is impossible, a tolerance band is recommended. We have considered the band to be $\pm 10\%$. The width of this band can be decided on the use case, narrow in cases where bias is not tolerable and broader where bias is not of much significance. The Four-Fifths Rule, Uniform Guidelines on Employee Selection Procedures (1978) of the US Government considers the adverse impact on a group if its selection rate is less than four-fifths (or 80%) of the rate of the group with the highest selection rate, thus indicating tolerance of 20% [50]. Accordingly, a Bias Index between 0 and 0.1 for a protected attribute indicates that the system is unbiased for that attribute. Similarly, the Fairness Score between 0.9 and 1.0 indicates a fair AI system.
10. Experiments on multiple datasets, some having bias and others balanced for bias, validate the proposed Fairness Score, the Bias Index and the SOP as these correctly assess the bias or fairness of the AI models.

10 Conclusion

Just as there are established procedures for security audit and certification of websites and portals by designated auditing agencies against accepted benchmarks such as OWASP Top 10, an ecosystem for fairness audit and certification for AI-based systems also needs to evolve. Standardization of the bias determining process is essential for more acceptance of such systems. This paper proposes a Fairness Score, Bias Index and a standard operating procedure to evaluate the fairness of supervised learning AI systems and provide a framework for Fairness Certification. Bias in an AI/ ML system can be present because of the training dataset, the algorithms used, or the way the model is used. These biases are measured using pre-defined metrics. A single metric may not be accurate for all the systems and applications, so a combination of metrics is preferred to evaluate their fairness. This paper outlines the desired features of such metrics and shortlists the suitable metrics for data-driven AI systems. As an AI/ ML system continuously learns while in operation, it is necessary to check for fairness periodically to ensure its neutrality and fair decision-making ability. Our experiments on the Census Income dataset, German Credit

dataset, and Health Insurance dataset establish that the Fairness Score and the SOP proposed in the paper can be used as a framework to evaluate the fairness of supervised learning AI systems and issue a Fairness Certificate.

Fairness Score and Bias Index are versatile as they provide a simple comparison of different AI systems on a linear scale. This work helps in fulfilling the ethical requirement of independent verification of fairness of AI systems and a reference scale for their comparison. Fairness Certificate generates a regulatory requirement and ensures trust, equity and transparency among the people. As governments use AI applications to deliver citizen-centric services, determining the fairness of such applications would become a legal requirement and citizens would be the beneficiaries of fairness certification. While procuring AI-based applications through a transparent tendering process, Fairness Certificate could be a qualification criterion. Developers would like to get their systems certified for fairness from third-party auditors for the wider acceptability of their products.

Further studies are suggested to extend the framework to specific use-cases and different types of AI systems. Some ethical questions arise as a result of the framework. Who decides the protected attributes of the dataset? Also, the selection of privileged and unprivileged classes can be manipulated to give a favourable result. There could be a bias or manipulation in deciding the protected attributes or the privileged/ unprivileged classes. Furthermore, a record might belong to a privileged class in one use case and an unprivileged class in another, making it difficult to have strict guidelines for privileged class selection. Answers to these concerns could be considered for further research and studies.

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